

Personalized AI assistants could either promote or hinder the emergence of collective intelligence

Jaeseung Kim¹ Sang Woo Park^{1,2,3,*}

¹ School of Biological Sciences, Seoul National University, Seoul, Korea

² Department of Ecology and Evolution, University of Chicago, Chicago, IL, USA

³ Department of Ecology and Evolutionary Biology, Princeton University, Princeton, NJ, USA

*Corresponding author: sangwoopark@snu.ac.kr

Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

27 Introduction

28 Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve
29 problems, make decisions, or adapt to changing environments exceeds each member’s
30 ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has
31 thus served as a central mechanism underlying the success of diverse organizations,
32 ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning
33 systems [10, 11]. Understanding the conditions that promote and maintain CI there-
34 fore remains a fundamental question across social sciences [12], evolutionary biology
35 [13], and computer science [14]. Addressing this question has become particularly
36 timely as artificial intelligence (AI) has begun to reshape how individuals acquire
37 information, make decisions, and contribute to collective outcomes [15].

38 The recent emergence of generative AIs and large language models (LLMs) has
39 changed the way individuals engage in various cognitive tasks, ranging from simple
40 everyday tasks to more complex tasks, including scientific discovery [16] and creative
41 writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has
42 been rapidly increasing [18], the potential long-term impact of personal AI services
43 on human society, especially the CI, remains uncertain. Some argue that AI can
44 promote CI by enhancing individual expertise [19], and opinion aggregation [20].
45 Conversely, empirical evidence indicate that the use of AIs can limit the diversity of
46 opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

47 Our ability to predict the long-term impact of personal AI services on human CI is
48 hampered by the apparent discrepancy in the incentives that shape AI uses. On one
49 hand, many industrial sectors are pushing towards increased AI usages, where the
50 number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On
51 the other hand, many academic institutions appear to discourage, and even penalize,
52 AI use, reflecting concerns about plagiarism, misinformation, and limited independ-
53 ence [28, 29]. Such discrepancy raises a question about whether directly penalizing
54 or incentivizing AI usage will benefit or harm human CI. These observations further
55 highlight the importance of indentifying a robust incentive structure that can pro-
56 mote CI in the presence of AI, which can retain the benefits of AIs while preventing
57 potential risk such as preserving the diversity of individual opinions.

58 To address these questions, we extended previous approaches for modeling CI,
59 where each individual makes independent predictions for the outcome from partial
60 observations of underlying causal factors [12, 13, 30]. We allowed each individual
61 to independently consult with personal AIs and revise their predictions accordingly.
62 Across different assumptions about the AI model’s ability and underlying incentive
63 structures, we investigate the robustness of human CIs.

Main

Model

To understand how AI affects CI, we start from a previous approach that models CI as a continuous-value prediction problem [13]. This model assumes that the prediction target is generated by a linear combination of independent causal factors $(x_1, x_2, \dots, x_M)$, and that each individual can observe only a single factor, hereafter referred to as interest, for their prediction. Collective predictions are obtained by aggregating individual predictions, and individuals evolve over time by imitating peers who receive higher payoffs under the incentive structure.

An individual (A) is characterized by three strategy variables: interest (e_A), belief (c_A), and AI belief (β_A). Individual predictions are made by multiplying the factor of interest by the individual's own belief:

$$Y_A = c_A x_{e_A} \quad (1)$$

Next, we represent AI as a prediction model with innate bias (b_i) and error (ϵ). We assume two extreme modes of AI based on how the AI responds to an individual user. The first is the “AI knows all” model, in which the AI predicts the target based on all factors regardless of the user's interest:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (2)$$

where α_i represents the true coefficient of the causal factor, and $\epsilon \sim N(0, \sigma_e^2)$ represents random error arising from the stochastic generative process of the AI. The second is the “AI answers question” model, in which the AI predicts the target based only on the user's interest:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (3)$$

Individuals receive assistance from the AI model and revise their predictions using a weighted sum of the AI prediction and their own prediction:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (4)$$

Each individual's payoff is then given by the expected value of their performance under the incentive structure. For each generation, two individuals are randomly sampled, and one imitates all three strategies of the other with a probability determined by the difference in expected payoff. Individual predictions are aggregated into a collective prediction by simple averaging in the AI knows all model, or by clustering in the AI answers question model, depending on the mode of AI.

AI can both promote or prevent the emergence of CI

To understand how AI could affect the emergence of CI, we first compare collective dynamics with and without AI assistance. We adopt two simple incentive structures, “feedback” and “niche expert” from Wang et al. [13] as a baseline, have been shown to promote near-perfect collective accuracy without AI (except averaging aggregation with niche expert incentive). The AI was parameterized to have moderate predictive performance, approximately 0.7.

We find that AI affects CI in qualitatively different ways depending on the mode of AI. In the AI knows all model, AI promotes CI by increasing both the speed at which CI emerges and the resulting collective accuracy (Fig. 1A). When the population can achieve near-perfect CI without AI (feedback incentive), AI accelerates its emergence by raising the initial collective accuracy to the level of AI accuracy. By contrast, when the population cannot achieve CI without AI (niche expert incentive), AI enables moderately accurate CI that is more accurate than either the AI or the human population can achieve alone.

However, in the AI answers question model, AI prevents the emergence and maintenance of CI regardless of the incentive structure (Fig. 1B). Under the feedback incentive, the population initially achieves near-perfect CI. However, as diversity of interest rapidly decreases, collective accuracy breaks down with an increase in collective variance. Under the niche expert incentive, the initial increase in collective accuracy peaks before near-perfect CI is achieved, and then plateaus with a slight decrease as collective bias increases.

A parsimonious model for RSV epidemics

We first began by asking whether a simple Susceptible-Infected-Recovered-Susceptible (SIRS) model can capture the observed RSV outbreak dynamics in Japan, including the sudden change in outbreak seasonality. The SIRS model is the simplest dynamical model that allows for the possibility of immune waning and therefore represents one of the most parsimonious models for explaining outbreak dynamics of respiratory infections. Here, we extended the standard SIRS model such that we could simultaneously estimate periodic seasonal transmission rates and non-periodic changes in transmission due to NPI measures that were implemented to prevent COVID-19 (Materials and methods).

Accounting for flexible changes in seasonal transmission rates that deviate from standard sinusoidal shapes allowed the SIRS model to reproduce the observed dynamics across all five islands, including changes in seasonality that occurred during 2016–2017 as well as post-pandemic changes in outbreak patterns (Fig. 2A). In contrast to most seasonal respiratory pathogens, which exhibit an annual cycle in transmission pattern, we estimated semiannual peaks in transmission rates in four islands: Honshu, Shikoku, Kyushu, and Ryukyu (Fig. 2B). These semiannual peaks in transmission rates were explained by the quadratic effects of specific humidity: in line with [?], we estimated that transmission would be maximized at a low and high mean

specific humidity and minimized at intermediate mean specific humidity (Fig. 2C). Interestingly, we found that minimum RSV transmission occurred at a much higher mean specific humidity in Ryukyu island than in other three islands (Fig. 2C). We did not find semiannual transmission rate patterns or quadratic humidity-transmission relationship for Hokkaido island (Fig. 2B–C); instead, we found that transmission decreased at very low specific humidity (< 5 g/kg). Differences in the ranges of observed mean specific humidity as well as the humidity-transmission relationship likely reflect heterogeneity in climate conditions. In particular, Honshu, Shikoku, and Kyushu islands are clustered around the main part of Japan, whereas Hokkaido and Ryukyu islands are located in the north and south from main islands, respectively (Supplementary Figure S2). Combining transmission rate estimates from all five islands still gave a quadratic humidity-transmission relationship for specific humidity between ≈ 5 g/kg and ≈ 18 g/kg, but the joint relationship poorly captured the humidity-transmission relationship in the Ryukyu island (Supplementary Figure S3).

We also found similar quadratic relationships between the estimated transmission rates and mean temperature (Supplementary Figure S4). Performing bivariate quadratic regressions against both humidity and temperature showed significant, positive effects of the quadratic humidity term in Honshu ($p = 0.01$) and Ryukyu ($p < 0.01$) islands (Supplementary Table S1). We found positive effects of the quadratic humidity term in Kyushu island but this effect was not significant ($p = 0.27$; Supplementary Table S1). We found negative effects of the quadratic humidity term in Shikoku island but this effect was almost close to zero and not significant ($p = 0.67$; Supplementary Table S1). However, we note that mean temperature and mean specific humidity are almost perfectly correlated, with correlation coefficients > 0.96 in all islands (Supplementary Figure S5), and therefore the bivariate regression cannot tease apart the effects of humidity and temperature separately.

We also compared R squared values from two separate quadratic regression models to test their ability to explain the variation in the estimated transmission rates using mean specific humidity and mean temperature as separate covariates (Supplementary Table S2). The humidity model outperformed the temperature model for Honshu (0.39 vs 0.29) and Ryukyu (0.59 vs 0.47) islands, whereas the temperature model outperformed the humidity model for Hokkaido (0.44 vs 0.52) and Shikoku (0.63 vs 0.72) islands. Both models had nearly identical R squared values for Kyushu islands (0.77 vs 0.77).

Across all five islands, we estimated large reductions in transmission during 2020 (Fig. 2D); however, there was large heterogeneity in the overall shape of the estimated NPI effects as well as the degree of transmission reduction. The reduction in transmission rates caused an increase in the susceptible pool (Fig. 2E), which allowed a large outbreak when NPIs were lifted (Fig. 2A). This observation is also consistent with [?] who predicted that an accumulation of the susceptible host population during the NPI period will eventually lead to a large outbreak.

Mechanisms for sudden changes in seasonality

Since the mechanistic SIRS model was able to accurately capture the observed transition in seasonality, we posit that it captures the relevant mechanisms for driving this pattern. Thus, we should be able to use the inferred dynamics to further tease apart the mechanisms underlying sudden changes in seasonality of the RSV outbreaks. To do so, we first began by evaluating the changes in the proportion of susceptible and infected individuals at the beginning of the season between 2013 and 2019. We focused on three islands where the sudden transition in RSV outbreak seasonality from winter to fall was observed: Honshu, Shikoku, and Kyushu islands.

Across three main islands, we found a consistent increase in the proportion of susceptible and infected individuals at the beginning of each season between 2013 and 2019 (Fig. 3A). These changes also corresponded with a decrease in center of gravity (Fig. 3A). A more detailed comparison of epidemic trajectories illustrated that an increase in the susceptible pool at the beginning of the season can drive a sudden shift in seasonality (Fig. 3B). The semiannual pattern in transmission (i.e., two peaks in transmission rates within a year) allows this transition, where a faster epidemic growth (from higher susceptible pool) drives a faster depletion of susceptibles and causes the epidemic to transition from a later peak to an earlier peak.

To understand why we observe a bigger shift in seasonal outbreak patterns in Kyushu island than in Honshu and Shikoku islands, we characterized how differences in the seasonal transmission patterns affects the difference in peak epidemic timing associated with an increase in population-level susceptibility (Fig. 3C–G). As a reference, we began with smoothed transmission rate estimates for Honshu (Fig. 3C) and Shikoku (Fig. 3F) islands and explore intermediate transmission patterns that interpolate two transmission patterns. Specifically, the transmission pattern in Kyushu exhibited a bigger amplitude and a wider trough between two transmission peaks. These differences were explored by varying the amplitude of the seasonal transmission rate (x -axis on Fig. 3G) and the degree of interpolation (y -axis on Fig. 3G), where the interpolation coefficients of 0 and 1 correspond to the seasonality in Honshu and Kyushu islands, respectively. Simulating the SIRS model using interpolated seasonal transmission rates showed that a large seasonal amplitude and wider trough cause larger changes in the timing of epidemic peak (Fig. 3G).

We did not observe a shift in RSV seasonality in Ryukyu island (Fig. 1E) despite the inferred quadratic humidity-transmission relationship (Fig. 2B). In Supplementary Materials, we show that there was limited variation in population-level susceptibility between 2014–2019 in Ryukyu island, explaining the lack of change in RSV seasonality (Supplementary Figure S6A). Likewise, we estimate a near constant susceptibility for 2013–2016 and 2019 for Hokkaido island and a small reduction in susceptibility during 2017 and 2018 (Supplementary Figure S7A).

Based on these observations, we tested whether an increase in transmission can also explain a sudden transition in RSV seasonality in Honshu, Shikoku, and Kyushu

islands (Supplementary Figure S8). This was done by estimating a separate seasonal transmission term before and after the change in RSV outbreak seasonality. We found that an increase in transmission was also able to capture the shift in RSV outbreak seasonality (Supplementary Figure S8A,B). Under this model, we estimated a moderate increase in mean transmission across three islands: 23% (95% CI: 19%–27%) for Honshu island, 16% (95% CI: 13%–18%) for Kyushu island, and 14% (95% CI: 7%–20%). The estimated shape of seasonal transmission remained largely unchanged (Supplementary Figure S8B).

Mechanisms for an increase in RSV transmission

The simple SIRS model predicted an increase in the proportion of susceptible individuals from 2013 to 2019 across Honshu, Shikoku, and Kyushu islands. As the SIRS model relies on simplifying assumptions about immunity against RSV infections, the estimated increase in the proportion of susceptible individuals likely reflects an effective increase in population-level susceptibility, rather than a strict increase in the raw number of susceptible individuals. This population-level susceptibility can be thought of as an average of how susceptible each person is to RSV infections across the population. Many factors can contribute to this population-level susceptibility, including immunological factors, such as the decreased probability of acquiring RSV following a primary RSV infection [?], and behavioral factors, such as increased exposure to RSV from increased contact rates. Alternatively, we found that an increase in transmission can also explain the sudden transition in RSV seasonality. Both models indicate an increase in the overall rate of RSV transmission ($\beta SI/N$) from 2013 to 2019. So what mechanisms caused the RSV transmission to increase over time in Japan?

Previously, [?] hypothesized that changes in climate and an increase in inbound overseas travelers may be both responsible for this shift in seasonality. While an increase in overseas travelers may also contribute to the increase in contact rates and therefore RSV transmission, it is likely to have weak effects given that the mean age of infection for RSV is typically very young. For example, a local surveillance effort in Kyoto reported that RSV infections were most frequently detected among < 6-year old (15.4% detection rate) compared to older age groups: 6–17 years (2.3%), 18–64 years (0.8%) and ≥ 65 years (0.6%) [?].

Alternatively, we hypothesized that the Comprehensive Support System for Children and Childcare, which was enacted in 2012 and launched in 2015, brought more children into the daycare system and thus increased the probability of exposure to RSV, leading to an increase in RSV transmission. This hypothesis builds on the work of [?] who previously suggested that increase in childcare attendance may have contributed to a large change in the seasonality of Kawasaki disease in Japan in the mid-2010s. An expansion of childcare facilities would increase contact rates among children, which in turn would increase the probability of exposure and therefore the population-level susceptibility against RSV infections. To quantify the potential

258 impact of this program, we compared the number of children attending childcare
 259 facilities in Japan since 2013 and compared them with our estimates of susceptible
 260 proportion (Fig. 4). Overall, we found consistent patterns of increase in childcare
 261 attendance and strong correlations with the estimated susceptible proportion: 0.977
 262 (95% CI: 0.847–0.997) in Honshu island, 0.943 (95% CI: 0.653–0.992) in Shikoku is-
 263 land, and 0.802 (95% CI: 0.124–0.970) in Kyushu island. The lack of island-level data
 264 did not allow us to test whether limited changes in susceptibility estimated for the
 265 Ryukyu island are correlated with a lack of increase in childcare facilities in Ryukyu
 266 island; however, counterfactual simulations show that an increase in susceptibility
 267 would have been able to shift the RSV outbreaks to spring in Ryukyu island, consis-
 268 tent with predictions for other islands based on the quadratic humidity-transmission
 269 relationship (Supplementary Figure S6B). Counterfactual simulations show that an
 270 increase in susceptibility would not be able to cause a sudden shift in the RSV
 271 outbreak seasonality in Hokkaido island, illustrating that the semiannual pattern in
 272 seasonal transmission is necessary to cause a sudden shift in outbreak seasonality.

273 Discussion

274 We present an epidemiological analysis of RSV outbreaks in Japan combining spa-
 275 tiotemporal observations with dynamical disease modeling. Our analysis revealed
 276 semiannual cycles in seasonal RSV transmission in four major islands (Honshu,
 277 Shikoku, Kyushu, and Ryukyu), which correlate with specific humidity and tem-
 278 perature. We found that these semiannual cycles allowed a sudden shift in the
 279 seasonality of RSV outbreaks in response to an increase in RSV transmission—this
 280 increase could be captured by either through changes in susceptibility or transmissi-
 281 bility. We hypothesize that an increase in childcare capacity through Comprehensive
 282 Support System for Children and Childcare may have contributed to the increase in
 283 RSV transmission [?].

284 Our analysis revealed considerable heterogeneity in epidemic dynamics across
 285 major islands in Japan. We showed that these differences could be explained by
 286 the differences in underlying seasonal transmission. Notably, we found a robust,
 287 quadratic relationship between the estimated transmission rates against mean spe-
 288 cific humidity and mean temperature across four major islands (Honshu, Shikoku,
 289 Kyushu, and Ryukyu), indicating low transmission at intermediate levels of specific
 290 humidity. These findings echo earlier studies that demonstrated similar relationships
 291 for RSV [?] and influenza [? ? ? ? ?]. However, given strong correlation between
 292 specific humidity and temperature, we were unable to identify the primary environ-
 293 mental driver from the current study. Their relative contributions may be separated
 294 from data spanning a wider geographical region with more climate variability as well
 295 as epidemiological variability [?].

296 The robustness of this quadratic relationship across islands exhibiting different
 297 climate conditions also suggests a possibility that climate-driven transmission may

298 be, in part, facilitated by human behavior: for example, an increase in time spent
 299 indoors during low and high humidity seasons can contribute to increased transmis-
 300 sion. Further investigation is needed to establish the underlying mechanism behind
 301 how climate conditions affect the transmission of respiratory pathogens, especially
 302 across different environmental contexts (e.g., indoor vs outdoor). We note that this
 303 relationship is correlational, rather than causal, and therefore any other climate vari-
 304 ables (e.g., rainfall) or seasonal variation in human behavior (e.g., school terms) that
 305 correlate with seasonal variation in specific humidity and temperature will be implic-
 306 itly captured by this relationship. Understanding why the relationship between RSV
 307 transmission and humidity in Hokkaido island differs from other locations remains
 308 to be answered.

309 We hypothesized that the Comprehensive Support System for Children and Child-
 310 care may have contributed to the increase in RSV transmission, but other mecha-
 311 nisms may have also contributed. For example, an increase in inbound overseas trav-
 312 elers may have led to an increased exposure to RSV among adults thereby increasing
 313 the total amount of transmission [?]. However, RSV typically infects young chil-
 314 dren [?] and contact rates among young children and adolescents (5–19) are much
 315 higher than among older adults in Japan [?], suggesting that an increase in
 316 inbound overseas travelers may have smaller effects than an increase in childcare at-
 317 tendance on overall RSV dynamics. Another competing hypothesis that could lead
 318 to an increase in transmission would be strain evolution: for example, one study
 319 noted that L172Q/S173L mutant strains of RSV B that became dominant around
 320 2016 had reduced susceptibility against monoclonal antibodies [?]. However, it is
 321 not yet clear how this mutation translates to susceptibility against infection-derived
 322 antibodies. While our findings are consistent with those by [?], who also pointed
 323 out the association between an increase in childcare attendance and a large change
 324 in the seasonality of Kawasaki disease in Japan, we cannot rule out the possibility
 325 that other factors, such as increase in air travel [?] or strain evolution [?], could
 326 have contributed to the increase in overall transmission.

327 Our sensitivity analysis revealed that an increase in transmissibility, rather than
 328 an increase in susceptibility, can also explain a sudden shift in seasonality in RSV
 329 outbreaks. In fact, both models provide evidence for increased contact rates. Specif-
 330 ically, the total rate of RSV transmission can be written as $\hat{\beta}cSI/N$, where $\hat{\beta}$ repre-
 331 sents the rate of transmission per contact, c represents the contact rate, S represents
 332 the number of susceptible individuals, I represents the number of infected individu-
 333 als, and N represents the total population size. Then, an increase in contact rate c
 334 can be captured by either increase in S (the original model) and $\hat{\beta}$ (the sensitivity
 335 analysis). Therefore, conclusions from both models are consistent with our hypoth-
 336 esis that an increase in contact rates, especially among children, contributed to the
 337 sudden change in RSV seasonality. Detailed age structured surveillance data, along-
 338 side age structured model, will be needed to properly assess the relative contribution
 339 of potential mechanisms, such as increase in childhood mixing patterns or increase
 340 in inbound overseas travelers.

Our model-based estimate of the initial susceptible fraction range between 8%–11%. However this estimate must be interpreted with care as it represents effective changes in population-level susceptibility, which can depend on many factors such as immunological and behavioral changes, rather than an actual susceptible fraction. Therefore, these estimates are not directly comparable to serological estimates of proportion seronegative. For example, [?] recently estimated that ~90% of individuals above 3 years of age have been infected at least once by analyzing anti-pre-F protein antibody concentration data from the Netherlands [? ?], but individuals with RSV antibodies can still permit reinfection. A cross-section serological assay from healthcare and non-healthcare workers at a pediatric medical facility in Japan found that ~95% of the participants were seropositive but only 63.5% had greater than 8 fold neutralizing antibody titers [?], suggesting challenges in translating serological data to susceptible fraction. While our estimates are not inconsistent with these data, detailed age structured serological data and model are needed to accurately estimate changes in population-level susceptibility against RSV.

Interventions to slow the transmission of COVID-19 have disrupted the circulation of many pathogens [? ? ? ?], including RSV epidemics in Japan. This disruption has added major challenges to predicting future outbreaks, which prevented us from making long-term predictions. Continued analysis of RSV dynamics in the post-COVID period, particularly with regard to whether RSV outbreaks in Japan return to fall or winter outbreaks, may help further validate our models.

Our analysis relied on several simplifying assumptions. For example, our model assumed that the waning of immunity can render previously infected individual to become fully susceptible to reinfections; this approximation allowed us to reconstruct the dynamics of susceptible hosts more easily. In practice, immunity is likely more complex with secondary infections being less susceptible and transmissible than primary infections [?]. Other studies have also suggested the importance of interaction between RSV A and B [? ?] as well as competition between RSV and human metapneumovirus [?]; our model did not account for such strain dynamics. We also did not account for explicit spatial structure or underlying stochasticity of the system. Therefore, our estimates of transmission rates must be interpreted with caution as they may implicitly capture factors that we did not account for explicitly, including strain dynamics, spatial structure, and exogenous transmission. Our analyses also primarily focused on three major islands (Honshu, Shikoku, and Kyushu), necessitating a better understanding of RSV dynamics in Hokkaido and Ryukyu islands; however, we note that these three islands make up > 95% of the population in Japan, meaning that our analyses capture the majority of RSV transmission in Japan. Despite these limitations, our model likely represents a parsimonious approximation of the complex host-pathogen interactions, allowing us to draw general conclusions about how interactions between endogenous (contact rates) and exogenous (climate-driven factors) factors can give rise to a sudden dynamical transition.

Understanding how endogenous and exogenous factors shape epidemic dynamics is critical to predicting future outbreaks and making public health decisions. Our

analysis shows that the interplay between climate-driven transmission and subtle changes in contact rates can cause a sudden transition in epidemic dynamics. More broadly, our analysis demonstrates that detailed epidemiological time series data can allow us to tease apart endogenous and exogenous factors in explaining dynamical transitions, offering unique insights into a long-standing ecological question.

Methods

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target Y is modeled as a linear combination of M independent factors $(x_1, x_2, \dots, x_m)$ and $M + 1$ coefficients $(\alpha_0, \alpha_1, \dots, \alpha_m)$, where α_0 denotes the global offset (Fig. 1A). The M factors are randomly drawn from normal distributions with mean X_i and standard deviation σ_i in each generation.

Among a population of N individuals, each individual can observe a single factor and make a prediction based on the observed value and their own belief. In particular, individual A 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (5)$$

where c_A represents individual A 's personal belief, e_A represents the factor that A observes (interest), and x_{e_A} represents the realized value of A 's interest. Individuals can also choose not to observe any factor, which represents stubborn individuals who make predictions only based on their own belief. For notational convenience, we assume that stubborn individuals have interest 0, and let x_0 always be 1.

Next, we model AI as a prediction system that has learned the true coefficients with bias and error (Fig. 1B). For each factor $i \in [0, M]$, AI knows the exact coefficient α_i and has a fixed bias b_i , resulting in a prediction of $\alpha_i x_i + b_i$. In terms of human-AI interaction, we model two contrasting modes of AI: the "AI knows all" model and the "AI answers question" model. In the AI knows all model, AI acts as an omniscient system and gives an answer regardless of the individual's interest, accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (6)$$

In contrast, the AI answers question model acts as a chatbot and gives an answer only based on the individual's interest. Thus, given the individual's interest, e_A , AI predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (7)$$

In both models, $\epsilon \sim N(0, \sigma_e^2)$ represents random error arising from the stochastic generative process of the AI model.

AI works as a personalized assistant for each individual, such that each individual receives assistance by incorporating the AI’s prediction. However, even with the same opportunity to receive AI assistance, reliance on AI can vary substantially across individuals. To model this, we assume that the revised prediction is a weighted sum of the individual’s own prediction and the AI’s prediction by introducing AI belief β_A as a weighting coefficient. The revised prediction of individual A can be written as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (8)$$

The revised predictions of individuals are aggregated by two different mechanisms, depending on the mode of AI. In the AI knows all model, even if an individual can observe only a single factor, their revised prediction contains information about all factors through interaction with AI. Therefore, we use simple averaging of revised predictions as the collective prediction. In contrast, in the AI answers question model, each individual gives a prediction based only on a single factor even after revising their prediction with AI assistance. Therefore, we use clustering aggregation to obtain the collective prediction, which is performed by clustering individuals based on interest and summing the average values of each cluster.

For each generation, each individual is rewarded with a payoff based on the given incentive structure (Fig. 1C). We adopt two incentive structures, “feedback” and “niche expert,” which were proposed by Wang et al. [13] and promote collective intelligence when prediction is performed without AI. The feedback incentive gives higher payoff to individuals who can push the collective prediction toward the true value, regardless of their individual performance. Given individual A ’s revised prediction $Y_A^{revised}$ and the collective prediction $\hat{Y}^{revised}$, it can be expressed as:

$$\pi_A = Y_A^{revised} (Y - \hat{Y}^{revised}) \quad (9)$$

The niche expert incentive gives higher payoff to individuals who have both better individual performance and a minority interest. Specifically, the niche expert payoff is expressed as the product of the proportion of individuals who observe the same factor as individual A , ρ_{e_A} , and individual accuracy. Given individual A ’s revised prediction $Y_A^{revised}$, it can be expressed as:

$$\pi_A = -\rho_{e_A} (Y_A^{revised} - Y)^2 \quad (10)$$

These two incentive structures were shown to achieve nearly perfect collective intelligence through social learning (except for the niche expert incentive with simple averaging aggregation) by promoting individual expertise while maintaining collective diversity. Therefore, we use these incentive structures as a baseline to understand how AI affects the emergence of collective intelligence. Moreover, we modify the two baseline incentives in a later section to understand the consequences of direct incentives and penalties for AI usage.

Throughout the simulation, individuals engage in social learning by imitating other individuals who receive higher payoffs. We assume that prediction events occur more frequently than social learning events within the population, such that individual payoff can be computed as an expected payoff. The imitation process is performed for at most one pair in each generation. By sampling two random individuals, A and B , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (11)$$

where s denotes the strength of social learning. When imitation occurs, A imitates all three strategies of B : interest, belief, and AI belief.

To evaluate and analyze the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, human accuracy, interest diversity, and median AI belief. These metrics capture the accuracy of the revised collective prediction, its bias–variance decomposition, the predictive performance without AI, the diversity of individual interests, and median AI belief in the population (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$	Squared systematic deviation of the revised collective prediction from the true value.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{revised}})$	Variability of the revised collective prediction error across prediction events.
Human accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the human-only collective prediction relative to the baseline prediction α_0 .
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median AI belief	$\text{median}_{A \in N}(\beta_A)$	Median value of AI belief β_A in the population.

465 Simulations

466 We run a series of simulations to understand how the interplay between climate-
 467 driven transmission and population-level susceptibility can drive a sudden shift in
 468 the timing of an epidemic. First, we simulated the model for a year (from week 26 of
 469 the starting year to week 25 of the following year) by varying the initial conditions
 470 and computing the center of gravity. Specifically, we varied $i(0)$ between 1×10^{-4}
 471 and 3×10^{-3} and $s(0)$ between 0.078 and 0.105. All other parameters were set to
 472 posterior median estimates.

473 To further understand how the shape of seasonal transmission term affects the
 474 degree of shift in seasonality, we varied the shape of seasonal transmission term by
 475 interpolating the estimated β_{seas} from Honshu and Kyshu islands, which are the two
 476 most populated islands. To do so, we first took posterior median estimates of β_{seas}
 477 from two islands and fitted generalized additive model with cyclic cubic spline bases
 478 to obtain smoothed estimates of β_{seas} for each island, which we denote as β_H and
 479 β_K , respectively. Then, we normalized seasonal transmission rates such that it has
 480 a mean of zero and has an amplitude of 1:

$$\zeta_X(t) = \frac{1}{\alpha_X} \left(\frac{\beta_X(t)}{\bar{\beta}_X} - 1 \right) \quad (12)$$

481 where $\zeta_X(t)$ represents the normalized seasonal transmission pattern in island X , and
 482 $\alpha_X = (\max(\beta_X(t)) - \min(\beta_X(t)))/2$ represents the amplitude of seasonal transmission
 483 pattern in island X . This allowed us to interpolate between two normalized seasonal
 484 terms and obtain a flexible shape for seasonal transmission rate:

$$\zeta_{\text{new}}(t) = \zeta_H(t)(1 - \theta) + \zeta_K(t)(\theta), \quad (13)$$

$$\beta_{\text{new}} = \left(\frac{\alpha_{\text{new}} \zeta_{\text{new}}(t)}{(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2} + 1 \right) \bar{\beta}_{\text{new}}, \quad (14)$$

485 where θ represents the interpolation coefficient, such that $\theta = 0$ and $\theta = 1$ causes
 486 β_{new} to have the same shape as β_H and β_K , respectively and $0 < \theta < 1$ allows
 487 us to model counterfactual transmission scenarios that interpolates between two
 488 islands. Note that $\zeta_{\text{new}}(t)$ does not necessarily have an amplitude 1 so we divide
 489 it by $(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2$ to ensure the amplitude of 1.

490 For a given value of the interpolation coefficient θ and seasonal amplitude α_{new} ,
 491 we simulated two outbreaks for a year assuming $s(0) = 0.078$ and $s(0) = 0.105$
 492 and computed the difference in the timing of epidemic peak. In doing so, all other
 493 parameters, including the mean transmission rate β_{new} , were fixed to posterior median
 494 estimates for the Honshu island.

495 Regression

496 We performed univariate quadratic regressions for the estimated transmission rates
 497 against mean specific humidity and mean temperature, separately, and computed R

squared values for each regression. We also perform bivariate quadratic regressions for the estimated transmission rates using both mean specific humidity and mean temperature as covariates. The significance of regression coefficients was determined using two-sided t-tests at the 0.05 significance level.

We performed a linear regression between the estimated proportion of susceptibles at the beginning of each season (week 26) against the number of children attending childcare facilities. For simplicity, the regression was performed using median estimates for the susceptible proportions. We did not have data on the number of children attending childcare facilities broken down by island level and so we used the national-level data instead.

Data availability

All data are stored in a publicly available GitHub repository (<https://github.com/parksw3/perturbation>) [?].

Code availability

All code are stored in a publicly available GitHub repository (<https://github.com/parksw3/perturbation>) [?].

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617 Contributions

618 S.W.P., I.H., and B.T.G. conceived of the study. S.W.P. performed the analysis and
619 wrote the initial draft. All authors (S.W.P., I.H., E.H., W.Y., R.E.B., G.A.V., S.C.,
620 C.J.E.M., and B.T.G.) reviewed and edited the manuscript.

621 Competing Interests

622 The authors declare no competing interests.

623 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ($n=52$). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ($n=52$). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ($n=8000$ posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ($n=8000$ posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $S(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$. (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from $n=8000$ posterior samples. Shaded regions represent the 95% credible interval in our estimates ($n=8000$ posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from $n=8000$ posterior samples. Error bars represent the 95% credible interval in our estimates ($n=8000$ posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.